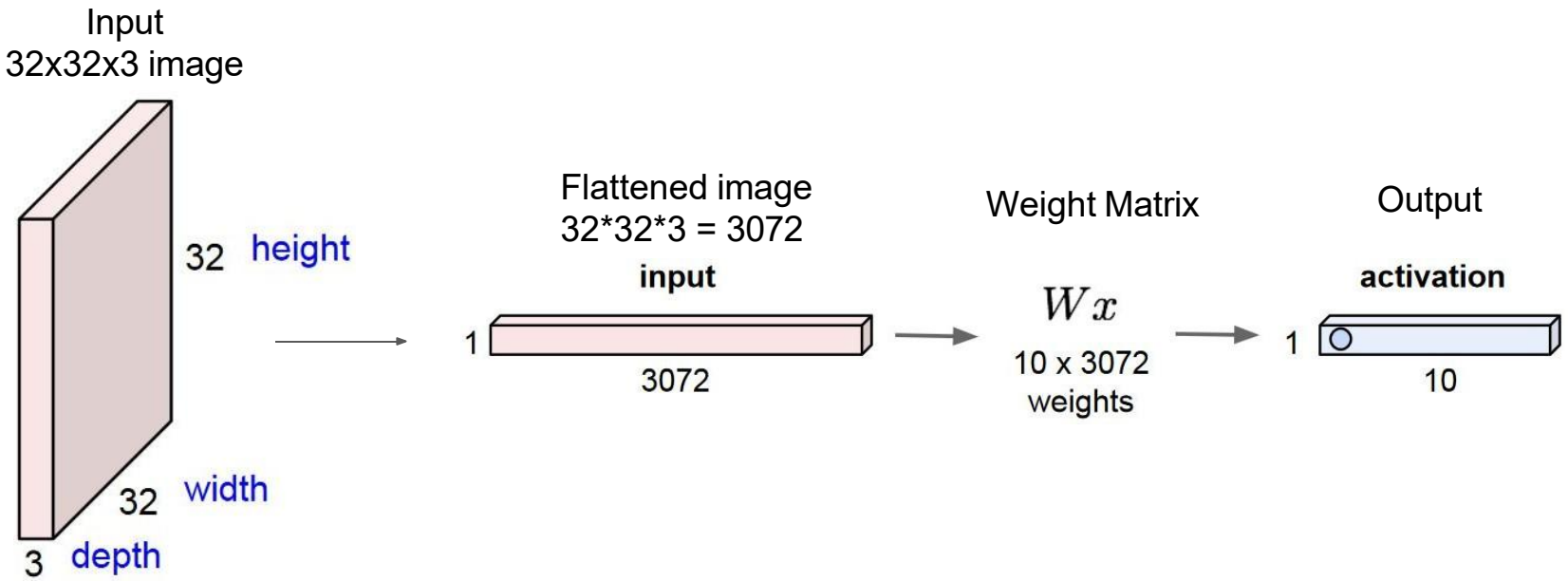
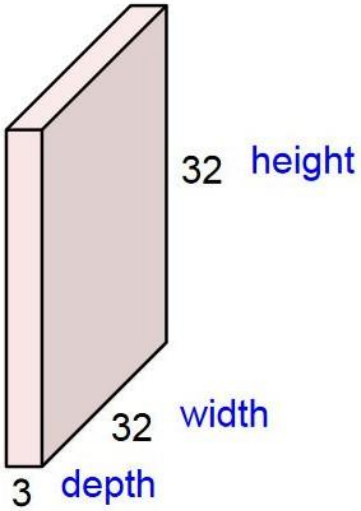
Fully Connected Layer

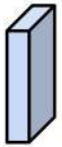


Convolutional Layer

Input
32x32x3 image



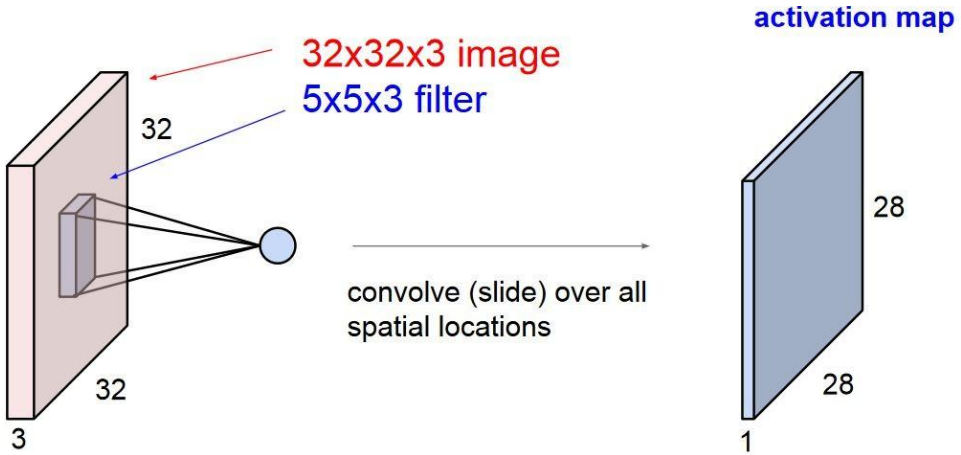
Filter
5x5x3



Convolve the filter with the image i.e. “slide over the image spatially, computing dot products”

Filters always extend the full depth of the input volume.

Convolutional Layer

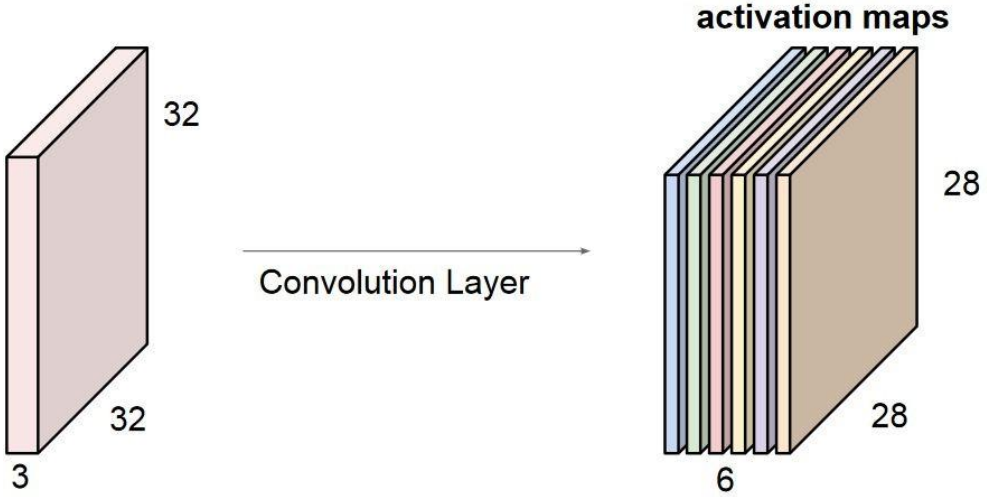


At each step during the convolution, the filter acts on a region in the input image and results in a single number as output.

This number is the result of the dot product between the values in the filter and the values in the 5x5x3 chunk in the image that the filter acts on.

Combining these together for the entire image results in the activation map.

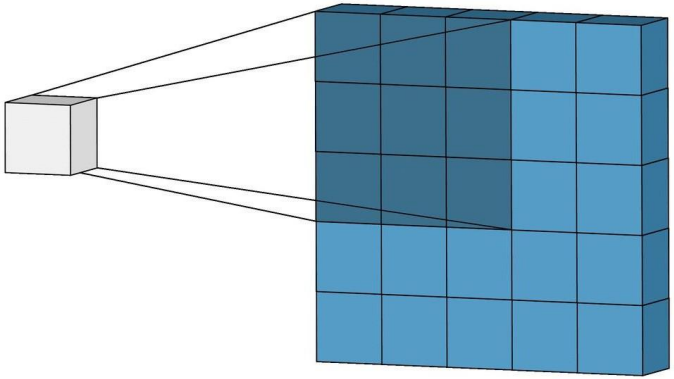
Convolutional Layer



Filters can be stacked together.

Example- If we had 6 filters of shape 5x5,
each would produce an activation map of 28x28x1 and our output would be a “new image” of shape 28x28x6.

Convolutional Layer

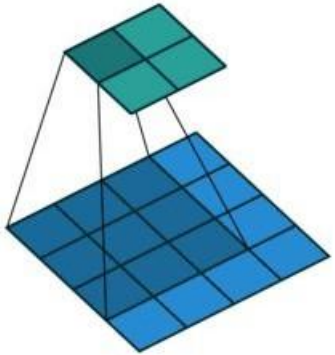


3_0	3_1	2_2	1	0
0_2	0_2	1_0	3	1
3_0	1_1	2_2	2	3
2	0	0	2	2
2	0	0	0	1

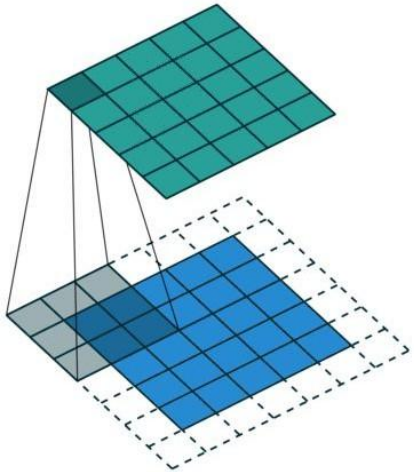
12.0	12.0	17.0
10.0	17.0	19.0
9.0	6.0	14.0

Visualizations borrowed from Irhum Shafkat's blog.

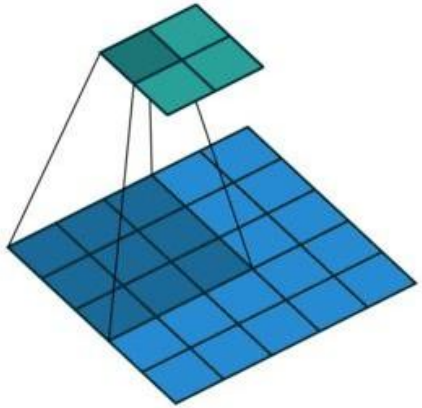
Convolutional Layer



Standard
Convolution



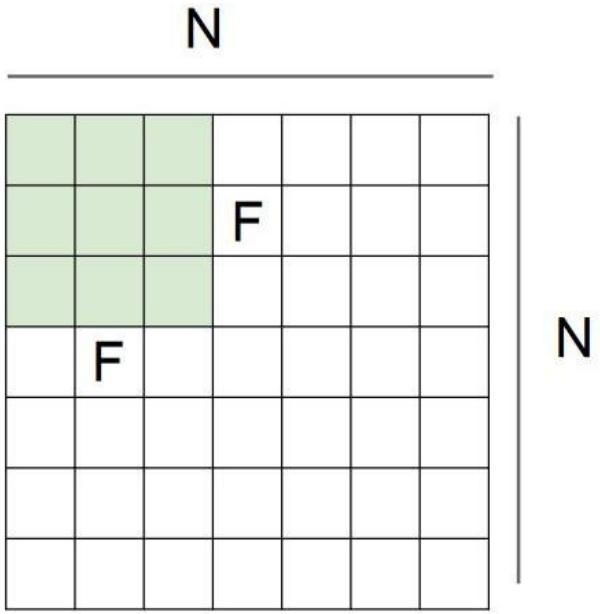
Convolution
with Padding



Convolution
with strides

Visualizations borrowed from vdumoulin's github repo.

Convolutional Layer

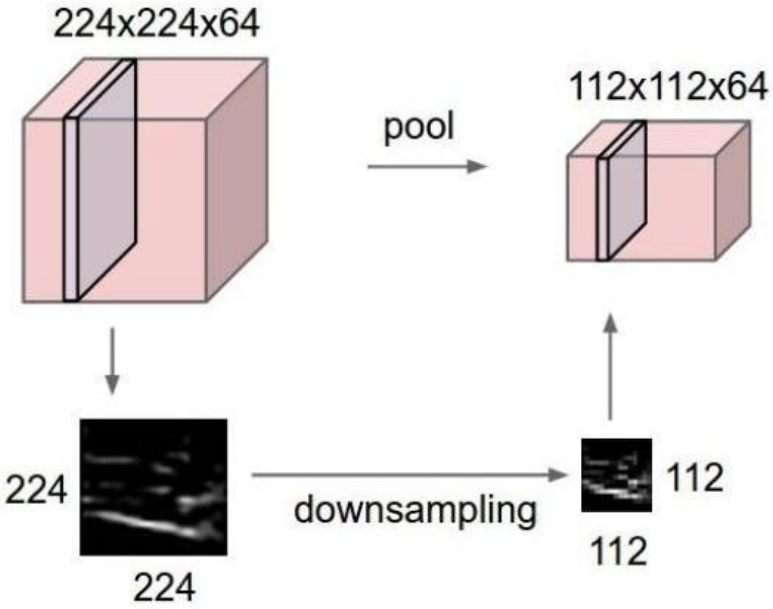


Output Size:
(N - F)/stride + 1

e.g. N = 7, F = 3, stride 1
=> (7 - 3)/1 + 1 = 5

e.g. N = 7, F = 3, stride 2
=> (7 - 3)/2 + 1 = 3

Pooling Layer



- makes the representations smaller and more manageable
- operates over each activation map independently